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Stable isotope analysis reveals ontogenetic feeding shifts in Pacific blue marlin (*Makaira nigricans*) off eastern Taiwan

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To gain a better understanding of the trophic ecology of Pacific blue marlin *Makaira nigricans* off eastern Taiwan, nitrogen and carbon stable isotopes ($\delta^{15}\text{N}$ and $\delta^{13}\text{C}$) and Bayesian mixing models were used to explore trophic dynamics and potential ontogenetic feeding shifts across *M. nigricans* of different size classes. *Makaira nigricans* samples from east of Taiwan ($n = 213$) and Palau ($n = 37$), as well as their prey ($n = 70$), were collected during 2012 and 2013. Results indicated increases in $\delta^{15}\text{N}$ with size, with values of larger size classes (> 200 cm eye-to-fork length; L_{EF}) significantly higher than those < 200 cm L_{EF} . Values of $\delta^{13}\text{C}$ were negatively correlated with size. *Makaira nigricans* > 200 cm L_{EF} had the highest estimated trophic position (4.44) and also exhibited ontogenetic changes in trophic position. Large *M. nigricans* fed more on dolphinfish *Coryphaena hippurus* and hairtail *Trichiurus lepturus*, while smaller *M. nigricans* consumed smaller forage fish (e.g., moonfish *Mene maculata*) and cephalopods. These changes may relate to greater swimming speeds and vertical habitat use in larger *M. nigricans*, allowing capture and consumption of larger prey items at higher trophic positions. The high trophic level of *M. nigricans* east of Taiwan confirms its important role as an apex predator in marine food webs and how ecological role changes with size.

KEYWORDS

apex predator, Bayesian mixing model, pelagic food web, size structure, trophic dynamics

1 | INTRODUCTION

Blue marlin *Makaira nigricans* Lacépède 1802 is a cosmopolitan species distributed throughout tropical, sub-tropical and temperate waters of the Pacific and Atlantic Oceans between 45° N and 45° S (Nakamura, 1985). Based on analyses of genetic and morphological data (Maguire *et al.*, 2006), it has been synonymised into a single species from two sibling species in the Indo-Pacific and Atlantic Oceans. Molecular genetic population analyses and fishery catch rates suggest a single stock of *M. nigricans* in the Pacific Ocean (Chen *et al.*, 2016; Graves & McDowell, 2003), though these data could neither disprove nor confirm the null hypothesis of panmixia among regional populations. Several stock assessments have been conducted for *M. nigricans*

in the Pacific Ocean, but the status of this population remains uncertain, ranging from healthy to fully exploited (Kleiber *et al.*, 2003; Lee *et al.*, 2014; Musyl *et al.*, 2015).

Makaira nigricans are pelagic apex predators that feed on a wide range of fishes and cephalopods (Nakamura, 1985). Feeding studies of *M. nigricans* around south-western Japan showed the most common prey species were skipjack tuna *Katsuwonus pelamis* (L. 1758) and dolphinfish *Coryphaena hippurus* L. 1758 and that larger *M. nigricans* fed primarily on larger prey (but less frequently) than smaller *M. nigricans*, which foraged more often but on a higher variety of prey species (Shimose *et al.*, 2006). Although stomach-content analyses (SCA) have provided information on the regional feeding habits of *M. nigricans*, SCA generally provides a snapshot of individuals, which makes it

difficult to compare feeding ecology across broad geographic and temporal scales (Estrada *et al.*, 2005). This is especially true for highly migratory fish such as tunas (Thunnidae) and billfishes (Istiophoridae and Xiphiidae), which are often only available for sampling over limited time periods and locations. Chemical tracer analysis of diet, such as nitrogen and carbon stable-isotope analysis (SIA), integrates diet over time and thus allows for trophic comparisons over longer timeframes (Graham *et al.*, 2010; Hussey *et al.*, 2010). Stable-isotope analysis has increasingly been used to address ecological questions over the past two decades (Madigan *et al.*, 2012; Rosas-Luis *et al.*, 2017).

As carbon stable-isotope ratios ($\delta^{13}\text{C}$) of animal tissues generally change little during the trophic transfer up food webs, they are used to evaluate the primary sources of energy (e.g., coastal–inshore carbon sources v. oceanic–planktonic sources of carbon sources) for organisms at various trophic levels. In contrast, nitrogen stable-isotope ratios ($\delta^{15}\text{N}$) have been demonstrated to increase predictably with increasing trophic level, providing means for quantitative estimates of an organism's trophic level (Caut *et al.*, 2009). Stable isotopes of carbon and nitrogen have been used successfully in freshwater and marine studies to elucidate not only trophic position, but also trophic shifts induced by disturbance and migration (Carlisle *et al.*, 2014; Davenport & Bax, 2002; Revill *et al.*, 2009). Applications of stable-isotope techniques to billfishes have generally focused on categorising billfish diet and trophic position (Ménard *et al.*, 2007; Rojas *et al.*, 2013; Tsai *et al.*, 2014).

To quantify the relative contribution of different prey sources to predator diets, stable-isotope studies often utilise mixing models that incorporate stable-isotope signatures from potential food sources, consumers and the change in stable-isotope values between predator and prey (trophic enrichment factor; TEF). By taking into account the error associated with these isotopic parameters, Bayesian mass-balance mixing models can quantitatively assess the relative contribution of different prey sources to the predator of interest (Madigan *et al.*, 2012; Phillips *et al.*, 2014; Shipley *et al.*, 2017). Mixing model estimates can be strengthened by incorporating additional information, such as gut content or faecal analyses to quantitatively or qualitatively refine the estimates of prey contributions to a consumer (Layman *et al.*, 2007; Varela *et al.*, 2017).

In this study, we conducted $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ isotope analysis on *M. nigricans* and prey muscle tissue to investigate feeding patterns, trophic changes with size and regional differences of *M. nigricans* from the north-western Pacific Ocean. We applied a Bayesian stable-isotope mixing model to estimate contributions of prey items to the diets of different size classes of *M. nigricans* from east of Taiwan. Our working hypothesis is that stable-isotope analyses may provide insights into not only the trophic ecology, but also the origin and migratory corridors of north-western Pacific *M. nigricans*. The presence of populations of *M. nigricans* in waters east of Taiwan and Palau with different isotopic signatures would provide evidence that the populations may have followed separate migratory pathways, possibly from different spawning locations. Ultimately, stable-isotope analysis provides the opportunity to better understand the role that apex *M. nigricans* play in the pelagic environment.

2 | MATERIALS AND METHODS

2.1 | Data collection and sampling

Makaira nigricans specimens were collected from east of Taiwan and Palau between January 2012 and January 2013 (Figure 1). *Makaira nigricans* were caught by commercial fisheries (e.g., longline and harpoon fisheries) and brought back to local fish markets for sale. The fishing area in waters off eastern Taiwan was from $\sim 21^{\circ} 27' - 23^{\circ} 37' \text{ N}$, $121^{\circ} 10' - 122^{\circ} 25' \text{ E}$. There were 37 specimens caught off Palau in August and the sampling area was $\sim 11^{\circ} 00' \text{ N}$, $132^{\circ} 00' \text{ E}$. Eye-to-fork length (L_{EF} ; posterior margin of bony orbit to the distal end of the central ray of the caudal fin; cm) and round mass (M_{R} , kg) were recorded for each fish in fish markets. White muscle samples were taken from the posterior hypaxial region, $\sim 10 \text{ cm}$ beneath the skin. Owing to its low lipid content and ease to with which it can be homogenised, white muscle tissue was found to be less variable in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ than all other tissues (e.g., red muscle tissue and liver tissue; Pinnegar & Polunin, 1999). Muscle samples were frozen at -80°C until processing. Muscle samples were acidified (10% HCl) to remove any residual carbonates, rinsed with distilled water and freeze-dried for 48 h.

Ethical research protocols and sampling guidelines were authorised under the auspices of the office of Director General, Fisheries Research Institute, Council of Agriculture, Executive Yuan, Keelung, Taiwan. Experiments were performed on frozen and chilled fish samples sourced from commercially regulated fisheries in Taiwan and Palau. On board commercial vessels, fish are quickly processed and frozen to minimise stress.

2.2 | Stable-isotope analysis

After freeze-drying, samples were ground to a homogeneous powder and ~ 0.5 to 0.7 mg of this powder was weighed in ultra-clean tin

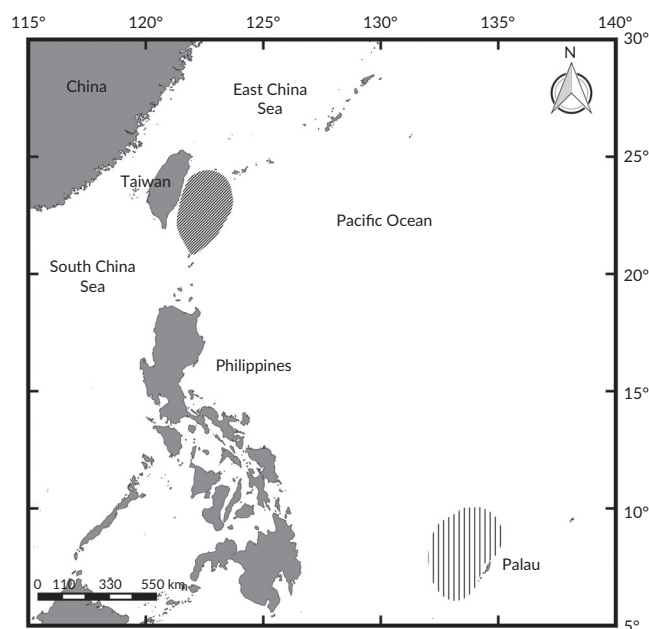


FIGURE 1 Sampling areas (▨) of *Makaira nigricans* in waters off eastern Taiwan and west of Palau from January 2012 to January 2013

capsules for stable carbon ($\delta^{13}\text{C}$) and nitrogen ($\delta^{15}\text{N}$) isotopic analyses (Davenport & Bax, 2002). Analytical precision (SD, $n = 5$) was 0.2‰ for both nitrogen and carbon, as estimated from standards analysed with the samples. Samples were combusted in an elemental analyser (Flash EA-1100 NC, Thermo-Finnigan; www.thermoscientific.com) to produce CO_2 and N_2 , which flowed through a gas chromatography (GC) column for separation into a mass spectrometer (Thermo Finnigan Delta^{plus} Advantage) for separately determining isotopic compositions. Isotopic values are expressed in parts per thousand (‰) as deviations from standards (Peedee belemnite limestone for $\delta^{13}\text{C}$ and nitrogen in air for $\delta^{15}\text{N}$):

$$\delta X = ((R_{\text{sample}} - R_{\text{standard}}))10^3,$$

where X is ^{13}C or ^{15}N and R is the corresponding $^{13}\text{C}:^{12}\text{C}$ or $^{15}\text{N}:^{14}\text{N}$ ratios (Peterson & Fry, 1987).

Food-web studies commonly examine the trophic dynamics and foraging relationships of marine organisms. The trophic position (T_P) of *M. nigracans* in each size class and their prey in waters off eastern Taiwan were calculated using the equation (Vander Zanden & Rasmussen, 1999): $T_P = ((\delta^{15}\text{N}_{\text{consumer}} - \delta^{15}\text{N}_{\text{baseline}})T_{\text{EF}}^{-1} + 2)$, where $\delta^{15}\text{N}_{\text{consumer}}$ represents the $\delta^{15}\text{N}$ value of *M. nigracans*, $\delta^{15}\text{N}_{\text{baseline}}$ represents the $\delta^{15}\text{N}$ value of the local baseline and T_{EF} represents the change in $\delta^{15}\text{N}$ between predator and prey. The $\delta^{15}\text{N}$ value of the herbivorous gastropod mollusc *Chlorostoma argyrostomum* in waters off eastern Taiwan were applied as the $\delta^{15}\text{N}$ baseline (6.24‰). The values applied were 2 for baseline and 2.4‰ for T_{EF} (Post, 2002). The T_{EF} of 2.4‰ was adopted for fish species, based on a meta-analysis of the literature for marine consumer diet $\delta^{15}\text{N}$ enrichment (Caut *et al.*, 2009; Olson *et al.*, 2010; Tsai *et al.*, 2015). The study of Caut *et al.* (2009) drew on $\delta^{15}\text{N}$ T_{EF} estimates that were lower than those previously reported and used previously to calculate trophic level; e.g., $3.40 \pm 0.27\text{‰}$, $n = 16$ (Minagawa & Wada, 1984), $3.40 \pm 0.13\text{‰}$, $n = 56$ (Post, 2002) and $3.21 \pm 0.2\text{‰}$, $n = 33$ (Robbins *et al.*, 2005).

2.3 | Size class and data analysis

Due to the retention of isotopes in consumer tissues in metabolic processes, the stable-isotope values increased with trophic level in food webs. Trophic discrimination factor (TDF) was the difference of stable-isotope values among predators, lower trophic level consumers and producers (Madigan *et al.*, 2012).

Makaira nigracans were grouped into three size classes that comprised: class I, 100–150 cm; class II, 151–200 cm; class III, > 200 cm. To test the significant differences of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values among area and body size, ANCOVA was used. Then, we used a Bayesian mixing model in MixSIAR (Parnell *et al.*, 2010) to estimate the relative contribution of prey diets to each size class. We grouped prey into seven general categories, including Carangidae, Bramidae, Scombridae, moonfish *Mene maculata* (Bloch & Schneider 1801), hairtail *Trichiurus lepturus* L. 1758, *C. hippurus* (< 60 cm) and Cephalopoda. Prey items were sampled in fish markets and they were caught by longline and set-net fisheries from commercial fishing in waters off eastern Taiwan. The tissue samples were processed by protocols as those for *M. nigracans*. The mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of prey groups were calculated and used as inputs in Bayesian mixing models. The

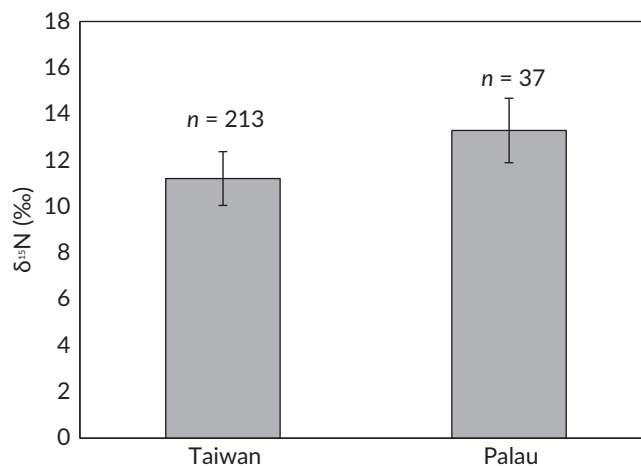


FIGURE 2 The $\delta^{15}\text{N}$ values (mean $\pm$ SD) of *Makaira nigracans* in Taiwan and Palau. Values of $\delta^{15}\text{N}$ off Palau were significantly higher than off Taiwan ($P < 0.05$)

M. nigracans specific trophic discrimination factor (TDF) used in mixing models was calculated using parameters from a study for western Pacific bluefin tuna *Thunnus orientalis* (Temminck & Schlegel 1844) where $\delta^{13}\text{C} = 1.1 \pm 0.5\text{‰}$ and $\delta^{15}\text{N} = 1.9 \pm 0.5\text{‰}$ (Caut *et al.*, 2009).

3 | RESULTS

3.1 | Sampling size of *M. nigracans* off eastern Taiwan and Palau

There were 213 samples from east of Taiwan and 37 samples from Palau. The size range of *M. nigracans* sampled in Taiwan was 109–249 cm and in Palau were 134–200 cm (mean $\pm$ SD $L_{\text{EF}} = 164.0 \pm 20.7$ cm). Range in mass was 22–149 kg (mean $\pm$ SD $M_R = 57.6 \pm 27.7$ kg).

3.2 | Stable-isotope analysis

Mean ($\pm$ SD) $\delta^{15}\text{N}$ of *M. nigracans* sampled in waters off Palau ($13.29 \pm 0.18\text{‰}$) was significantly greater than Taiwan ($11.21 \pm 1.16\text{‰}$; $P < 0.05$; Figure 2). Values of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ in *M. nigracans* showed significant differences between Taiwan and Palau (ANCOVA, $\delta^{15}\text{N}$: $F_{1, 246} = 145.20$, $P < 0.05$, $\delta^{13}\text{C}$: $F_{1, 246} = 68.19$, $P < 0.05$). Body size of *M. nigracans* from east of Taiwan was positively and significantly correlated to $\delta^{15}\text{N}$ but was negatively correlated to $\delta^{13}\text{C}$ ($P < 0.05$; Figure 3a). Correlations between size and isotope values ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) from Palau samples were non-significant ($P > 0.05$; Figure 3b). Correlations between $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values in samples from Taiwan and Palau were non-significant (Figure 4). Isotope values for *M. nigracans* collected from east of Taiwan and Palau by size class are given in Table 1. For Taiwan samples, the $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ values in size-class III were significantly higher than other size classes whereas no significant differences were observed between size classes in samples from Palau.

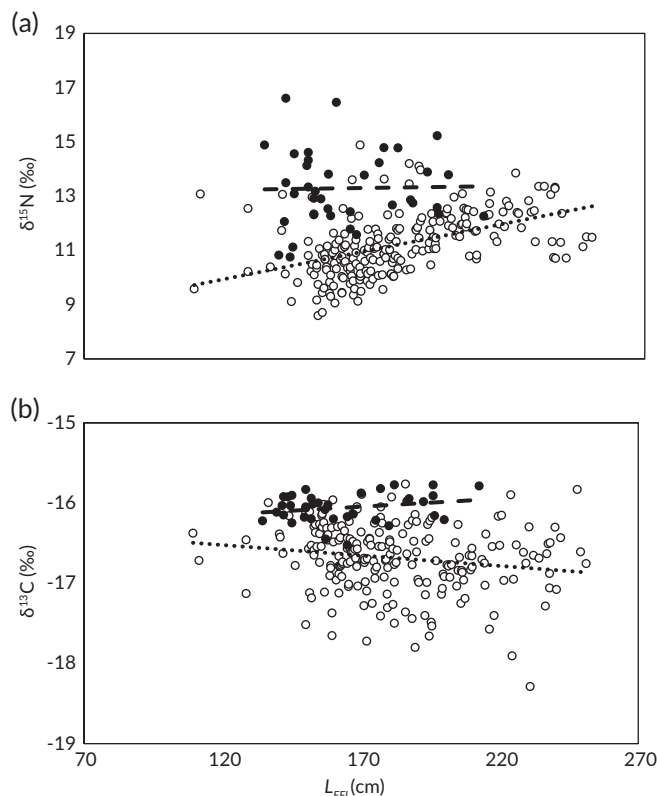


FIGURE 3 The relationships of a $\delta^{15}\text{N}$ and b $\delta^{13}\text{C}$ values with eye-fork length (L_{EF}) of *Makaira nigricans* off Taiwan ($n = 213$; $\delta^{15}\text{N}$: $y = 0.02x + 7.4857$, $R^2 = 0.2296$; $\delta^{13}\text{C}$: $y = -0.0026x - 16.213$, $R^2 = 0.0288$) and Palau ($n = 37$; $\delta^{15}\text{N}$: $y = 0.0014x + 13.069$, $R^2 = 0.0004$; $\delta^{13}\text{C}$: $y = 0.002x - 16.387$, $R^2 = 0.0546$). (○) Taiwan ($n = 213$), and (●) Palau ($n = 37$)

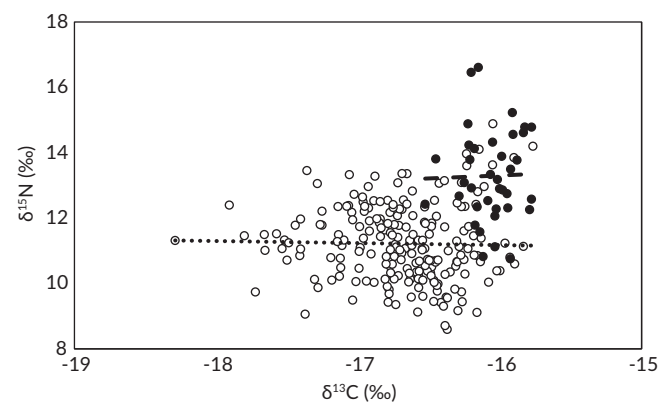


FIGURE 4 The relationships between the $\delta^{15}\text{N}$ value and $\delta^{13}\text{C}$ value of *Makaira nigricans* off Taiwan ($n = 213$; $y = -0.0632x + 10.16$, $R^2 = 0.0005$; $P > 0.05$) and Palau ($n = 37$; $y = 0.1842x + 16.25$, $R^2 = 0.0005$; $P > 0.05$). (○) Taiwan ($n = 213$), and (●) Palau ($n = 37$)

3.3 | Trophic level

A total of 70 prey taxa were collected from east of Taiwan. Most prey items were Carangidae, Scombridae, Bramidae, Cephalopoda, *M. maculata*, *T. lepturus* and *C. hippurus* (Table 2). Stable-isotope biplots of *M. nigricans* in each size class and their prey items are shown in Figure 5. Samples from east of Taiwan showed higher variability in both $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$. Class III *M. nigricans* from east of Taiwan

showed the highest $\delta^{15}\text{N}$ value and trophic level (12.10‰ and 4.44), followed by class II (10.92‰ and 3.95) and class I (10.98‰ and 3.97). Across prey groups, *T. lepturus* showed the highest mean $\delta^{15}\text{N}$ value and thus highest estimated trophic position (12.80‰ and $T_p = 4.73$), followed by *C. hippurus* (9.42‰ and $T_p = 3.32$) and Scombridae (9.10‰ and $T_p = 3.19$; Figure 6).

3.4 | Stable-isotope mixing model

The Bayesian mixing model indicated prey items were different among each size class. Important prey items for samples from east of Taiwan in class I were *M. maculata* (30.2%), Cephalopoda (23%), Carangidae (13.7%), *C. hippurus* (10.9%). The dominant prey items in class II off eastern Taiwan were *M. maculata* (29.7%), followed by Cephalopoda (22.2%), *C. hippurus* (16%) and Carangidae (10.4%). The dietary contributions of Bramidae and *T. lepturus* were the lowest in class I and II (Bramidae: 4.3% for class I and 6.6% for class II; *T. lepturus*: 7.8% for class I and 7.1% for class II). The dietary contributions of *C. hippurus* (45.9%) and *T. lepturus* (27.8%) became higher in class III, more than 70% and the contributions of other prey items were lower by 10% (Table 2).

4 | DISCUSSION

4.1 | Size class

We used stable-isotope analysis and Bayesian mixing models to estimate trophic position and trophic dynamics across ontogeny of *M. nigricans* off eastern Taiwan and Palau. The results indicated significant, positive correlation between body size and $\delta^{15}\text{N}$ values. These results show size-based changes in the role of *M. nigricans* in pelagic ecosystems in the western Pacific Ocean and demonstrate the utility of stable-isotope analysis to better understand the ecology of *M. nigricans* and other large pelagics in the region.

Many marine migratory species have displayed similar patterns of increasing trophic position with size, with larger individuals showing higher $\delta^{15}\text{N}$ values presumably due to the ability of larger fish to capture larger and more diverse prey at higher trophic levels (Davenport & Bax, 2002; Graham et al., 2007; Tsai et al., 2014). Different body sizes of predators have also been linked to differences in resource utilisation and shifts in preferred habitat (Werner & Gilliam, 1984). For example, Tsai et al. (2014) found increasing $\delta^{15}\text{N}$ values in larger sailfin *Istiophorus platypterus* (Shaw 1792) off eastern Taiwan, which paralleled the results of *M. nigricans* indicating a commonality in ontogenetic foraging patterns. At body sizes <180 cm, *I. platypterus* fed mainly on lower trophic-level prey items such as red bigeye *Priacanthus macracanthus* Cuvier 1829 and *Auxis* spp. and then shifted to higher value prey such as *T. lepturus* at larger sizes (> 180 cm). Both of these studies indicate that *T. lepturus* are an important prey resource for billfishes off eastern Taiwan. In contrast to *M. nigricans* in Taiwan, those in Palau showed no isotopic increase with body size. Isotopic studies on *T. orientalis* in western and eastern Pacific showed similar patterns to our study in that $\delta^{15}\text{N}$ values of *T. orientalis* in the western Pacific Ocean increased with size while those in eastern

TABLE 1 Values of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ in different size classes of *Makaira nigricans*

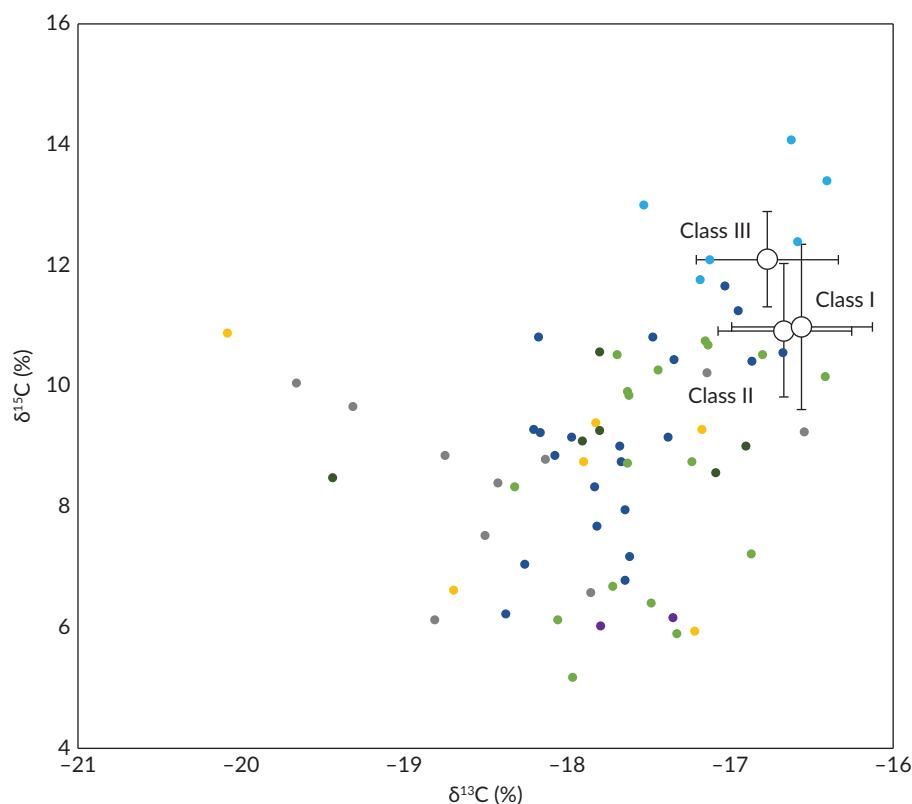
Samples	<i>n</i>	<i>L</i> _{EF} (cm)	δ ¹⁵ N (‰)				δ ¹³ C (‰)			
			Mean	SD	Minimum	Maximum	Mean	SD	Minimum	Maximum
Taiwan										
Class I	13	100–150	10.98	1.36	9.11	13.08	−16.56	0.43	−17.52	−16.00
Class II	148	151–200	10.92	1.11	8.59	14.88	−16.67	0.41	−17.80	−15.77
Class III	52	> 200	12.10	0.80	10.68	13.85	−16.77	0.44	−18.29	−15.83
Palau										
Class I	13	100–150	13.37	1.77	10.75	16.61	−16.05	0.13	−16.25	−15.83
Class II	24	151–200	13.25	1.19	11.58	16.46	−16.06	0.20	−16.53	−15.78

L_{EF} , Length from eye to tail fork.

TABLE 2 Values of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ for prey species of *Makaira nigricans* in Taiwan used in this study. 5% and 95% indicated the confidence interval from 5% to 95%

Prey item	Class I			Class II			Class III		
	Mean (‰)	5% CI	95% CI	Mean (‰)	5% CI	95% CI	Mean (‰)	5% CI	95% CI
Bramidae	4.3	0	16.6	6.6	0	21.4	4.2	0	14.4
Carangidae	13.7	0	65	10.4	0.1	47.5	3.1	0	13.3
Cephalopoda	23	0	89.3	22.2	0.1	74.7	8.4	0	35.7
<i>Coryphaena hippurus</i>	10.9	0	39.4	16	0.1	47.1	45.9	23.5	68.5
<i>Mene maculata</i>	30.2	0	80.3	29.7	0	71.1	5.6	0	21.6
Scombridae	10.2	0	67.8	7.9	0.1	30.9	4.9	0.1	18.5
<i>Trichiurus lepturus</i>	7.8	0	23.5	7.1	0	18.4	27.8	16.8	38.2

Class I, 100–150 cm eye-to-tail fork length (L_{EF}); class II, 151–200 cm L_{EF} ; class III, > 200 cm L_{EF} .

**FIGURE 5** Stable-isotope ($\delta^{15}\text{N}$ and $\delta^{13}\text{C}$) biplot of *Makaira nigricans* in different size classes (O: class I, 100–150 cm eye-to-tail fork length (L_{EF}); class II, 151–200 cm L_{EF} ; class III, > 200 cm L_{EF}) and their prey items in Taiwan. (●) Carangidae, (●) Bramidae, (●) Scombridae, (●) Cephalopoda, (●) *Trichiurus lepturus*, (●) *Mene maculata*, and (●) *Coryphaena hippurus*

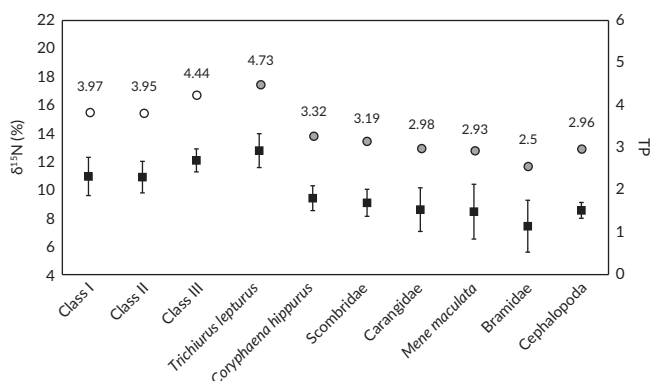


FIGURE 6 The trophic position (numerical values) and estimated from mean ($\pm$ SD) $\delta^{15}\text{N}$ values (■) of *Makaira nigricans* in each size class (O; class I, 100–150 cm eye-to-tail fork length (L_{EF}); class II, 151–200 cm L_{EF} ; class III, > 200 cm L_{EF}) and the prey items (●) off Taiwan

Pacific Ocean did not (Madigan *et al.*, 2012, 2016). This discrepancy may be related to the differences of primary production and food-web structure in different regions (Sommer *et al.*, 2002). Ontogenetic shifts in stable isotopes of several pelagic species showed variations on different regional scales.

Values of $\delta^{13}\text{C}$ represent the carbon sources from primary producers and in marine systems mirror the spatial and temporal variability of the environment. In particular, $\delta^{13}\text{C}$ tends to decrease in open pelagic waters, with increasing distance from coastlines (von Biela *et al.*, 2016). *Makaira nigricans* $\delta^{13}\text{C}$ values were negatively correlated with body size, with sizes > 200 cm showing the lowest $\delta^{13}\text{C}$ values. These patterns have several plausible causes. Larger individuals seem to feed on prey items with lower $\delta^{13}\text{C}$ values, potentially suggesting higher use of more pelagic resources and less coastal prey items. Related to this feeding shift, electronically-tagged large individuals exhibited long distance migratory behaviours that may traverse more oligotrophic habitats with lower baseline $\delta^{13}\text{C}$ values. Previous studies indicated that other pelagic species, such as salmon sharks *Lamna ditropis* Hubbs & Follett 1947, shifted from coastal juvenile habitats to oceanic habitats when they matured (Carlisle *et al.*, 2012; Weng *et al.*, 2007). Werner & Gilliam (1984), indicated foraging ecology changes through ontogeny and that shifts in position within food webs and are reflected by lower $\delta^{13}\text{C}$ values.

4.2 | Trophic structure

Bayesian mixing models estimated that the diet compositions of *M. nigricans* shifted by size class east of Taiwan. Trophic position of *M. nigricans* ranged from 3.97 for smaller fish (< 200 cm) to 4.97 for larger fish (> 200 cm). The increase of $\delta^{15}\text{N}$ values and increased trophic position has been correlated to increases in body mass in large fish (Lindsay *et al.*, 1998; Jennings *et al.*, 2002) and many pelagic predators display ontogenetic shifts in diet and resultant higher trophic position changes with size (Estrada *et al.*, 2006; Madigan *et al.*, 2016). Similar patterns were found in yellowfin tuna *Thunnus albacares* (Bonnaterre 1788), with higher $\delta^{15}\text{N}$ values observed at sizes > 45 cm in Ecuador (Varela *et al.*, 2017) and changing diet compositions at different growth stages (Popp *et al.*, 2007). The important prey items for

small *M. nigricans* (100–150 cm) were smaller forage prey (*M. maculata* and cephalopods) and small *C. hippurus*, while the dominant prey items for large fish (> 200 cm) were *C. hippurus* and *T. lepturus*. Previous studies indicated that pelagic predators display a wide range of trophic positions throughout different life stages concurrent to the feeding abilities associated with body size (Revill *et al.*, 2009). For example, large swordfish *Xiphias gladius* L. 1758 are able to exploit more resources via extended use of the vertical water column, consuming larger prey resources inhabiting deeper meso- and bathypelagic regions (Ménard *et al.*, 2007). It is possible that larger *M. nigricans* are more capable of diving to and below the thermocline for limited periods to exploit larger, higher trophic level prey.

4.3 | Geographic effects of stable isotope

Makaira nigricans from Palau had higher $\delta^{15}\text{N}$ values and estimated trophic positions than those from east of Taiwan, though their body sizes were smaller than samples from Taiwan. Stable-isotope analysis can provide information of regional foraging ecology and different values of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ in the same species may reflect regional differences due to distinct oceanographic–environmental factors such as the abundance of primary producers or the strength of nitrogen fixation process (Graham *et al.*, 2010). The low $\delta^{15}\text{N}$ values of *M. nigricans* in east of Taiwan may be due in part to the oligotrophic Kuroshio Current. By contrast, localised upwelling caused by internal waves may drive more localised upwelling and consequently higher $\delta^{15}\text{N}$ baseline values in Palau (Wolanski *et al.*, 2004). Madigan *et al.* (2016) showed a similar pattern in *T. orientalis*, in that $\delta^{15}\text{N}$ values in east of Taiwan were lower than those in the more productive waters off Japan and in the Kuroshio–Oyashio transition region.

A variable range of trophic levels of *M. nigricans* east of Taiwan was documented in this study. Foraging shifts by body size were revealed, providing insights into the ontogenetic shifts in the trophic dynamics of apex predators in pelagic ecosystems. Though it is difficult to obtain *M. nigricans* stomachs for traditional stomach-content analysis, diet studies could corroborate or refute the SIA-based diet inferences provided here. Further holistic investigations need to focus on other components of the pelagic ecosystem in Taiwan, including primary production (phytoplankton and zooplankton), billfish and sharks (and their prey items). The variability and importance of this ecosystem in relation to the Kuroshio pelagic food web structure will provide insights into overall pelagic fisheries productivity in the Northwest Pacific Ocean.

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